

JKBOSE 2026

CLASS 11TH

#XUV 

CHEMICAL EQUILIBRIUM

LECTURE-3 CHEMISTRY

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ON
TOP.



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Topics To Covered

- 1** Relation Between K_p and K_c
- 2** Applications of Equilibrium Constants

REVISION

CHEMICAL EQUILIBRIUM.

Dynamic
Nature of
Equilibrium

Law of mass
Action



$$\text{Rate} = k[A]^a[B]^b$$

↓

Law of Chemical Equi.

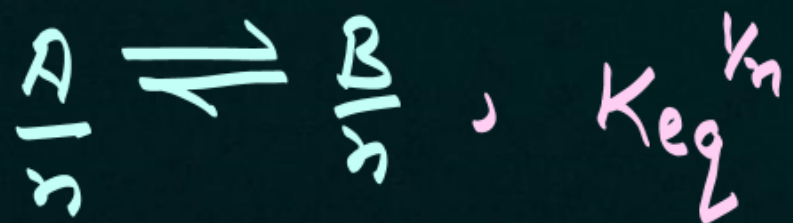
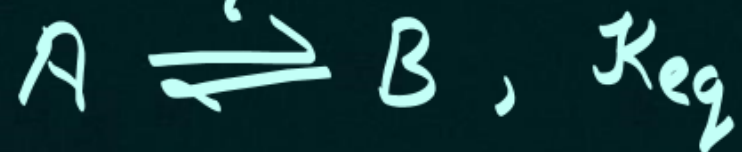


$$\frac{[C]^c[D]^d}{[A]^a[B]^b} = K_{eq}$$

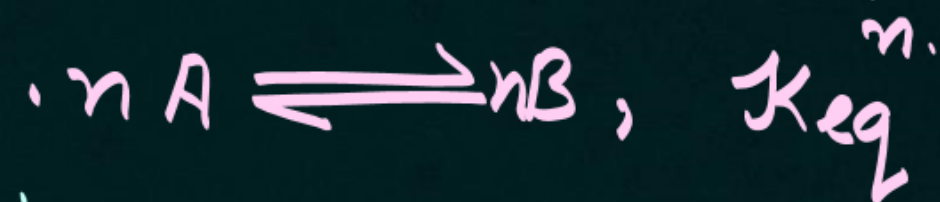
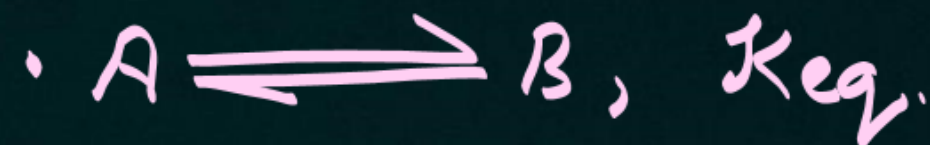
Characteristics of Equilibrium Constant $\Rightarrow$

1) The value of K_{eq} does not depend upon concentration and only depends upon the temperature.

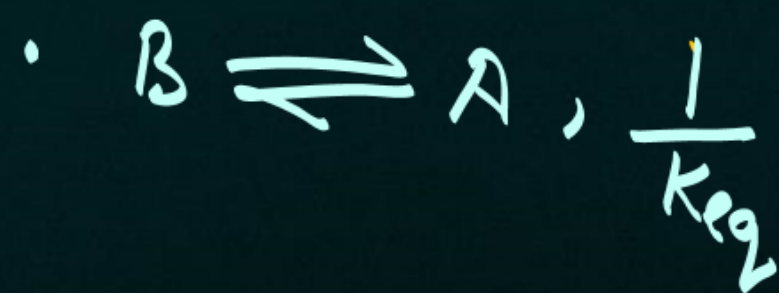
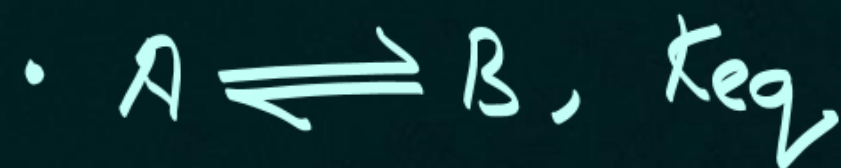
2) If the equilibrium reaction is divided by n , then, new K_{eq} will



3) If the equilibrium reaction is multiplied by n , then equilibrium constant changes to.

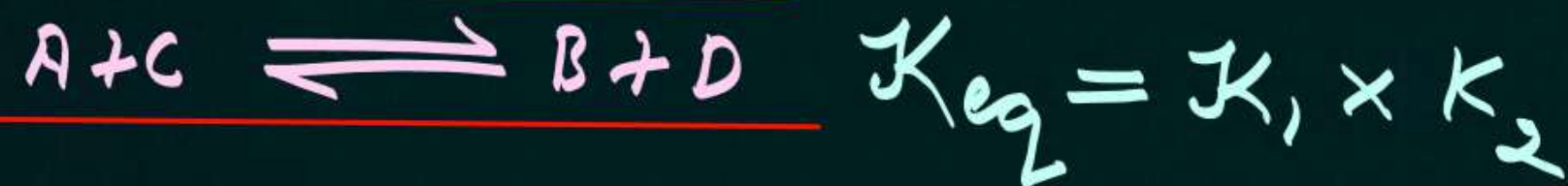


4) If the equilibrium equation is reversed, then new equilibrium constant will be.

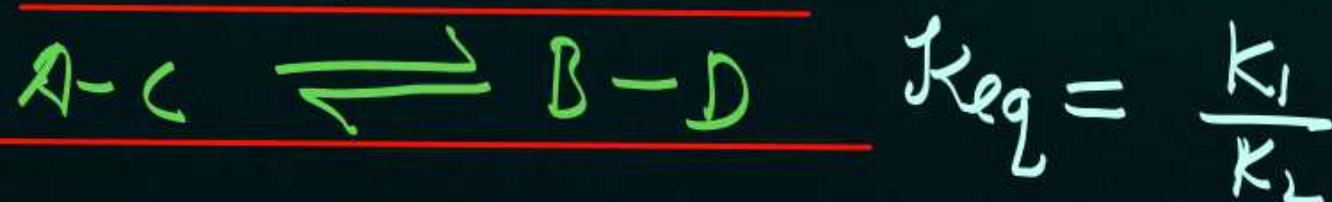


REVISION

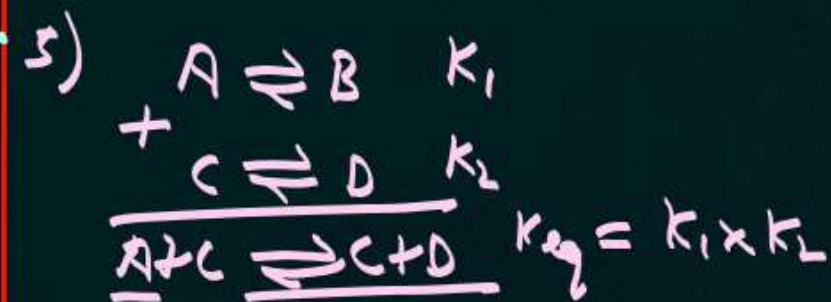
5) If two equilibrium equations are combined then, new K_{eq} will be.



6) If two equilibrium equations are subtracted then, new K_{eq} will be



1) K_{eq} is indep. of conc. and depend on temp.



K_p & K_c .

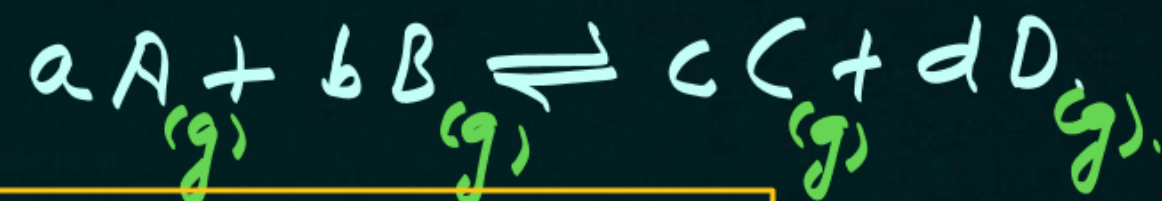
K_{eq} is expressed in two ways.

1) $K_c \rightarrow$ (Concentration constant). It is expressed when substances are in solutions or gas phases. Concentration is expressed in terms of molarity (mol l^{-1}).

e.g. $aA + bB \rightleftharpoons cC + dD$.

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

2) $K_p \rightarrow$ (Pressure constant). It is expressed when substances are in gas phase. Term partial pressure is used to express the amount. i.e.



$$K_p = \frac{P_c^c P_d^d}{P_A^a P_B^b}$$

A, B, C, D.

$$P_{\text{Total}} = P_A + P_B + P_C + P_D$$

Individual gas press = Partial pressure

Relation Between K_p and K_c

Consider a gas phase equilibrium reaction



Then, according to law of Chemical equilibrium.

$$K_p = \frac{P_C^c P_D^d}{P_A^a P_B^b}$$

→ (1)

According to ideal gas equation, $PV = nRT$.

$$P = \frac{n}{V} RT.$$

$$P = CRT$$

$$\text{i.e. } P_C^c = (C_C RT)^c$$

$$P_D^d = (C_D RT)^d$$

$$P_A^a = (C_A RT)^a$$

$$P_B^b = (C_B RT)^b$$

Put these value in equation (1).

$$K_p = \frac{(C_C RT)^c (C_D RT)^d}{(C_A RT)^a (C_B RT)^b}$$

$$K_p = \frac{(C_C RT)^c (C_D RT)^d}{(C_A RT)^a (C_B RT)^b}$$

$$K_p = \frac{C_C^c C_D^d (RT)^c (RT)^d}{C_A^a C_B^b (RT)^a (RT)^b}$$

$$K_p = \frac{C_C^c C_D^d (RT)^{c+d}}{C_A^a C_B^b (RT)^{a+b}}$$

$$K_p = \frac{C_C^c C_D^d}{C_A^a C_B^b} (RT)^{(c+d)-(a+b)}$$

$$K_p = \frac{C_C^c C_D^d}{C_A^a C_B^b} (RT)^{n_p - n_R}$$

n_p = no. of moles of products (g)
 n_R = no. of moles of reactants (g)

$$K_p = \frac{C_C^c C_D^d}{C_A^a C_B^b} (RT)^{\Delta n_g}$$

$$K_p = \frac{[C]^c [D]^d}{[A]^a [B]^b} (RT)^{\Delta n_g}$$

$$K_p = K_c (RT)^{\Delta n_g}$$

Units of Equilibrium Constant.

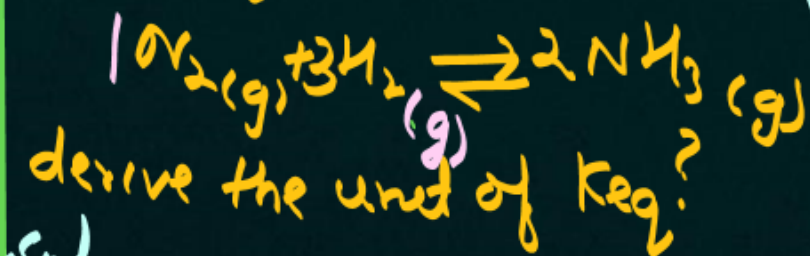
1) If $\Delta n_g = 0$ ($n_p - n_r = 0$) or no. of moles of product becomes equal to the no. of moles of reactants then K_{eq} has no unit.

2) But Δn_g is not equal to zero, the unit of K_{eq} is derived from following relation.

$$K_c = (\text{mol l}^{-1})^{\Delta n_g}$$

or $K_p = (\text{atm})^{\Delta n_g}$

Q. For equilibrium $\Rightarrow$



sol $K_{eq} = (\text{mol l}^{-1})^{\Delta n_g}$

$$\begin{aligned} K_c &= \frac{[C]^c [D]^d}{[A]^a [B]^b} = \frac{(\text{mol l}^{-1})^c (\text{mol l}^{-1})^d}{(\text{mol l}^{-1})^a (\text{mol l}^{-1})^b} \\ &= \frac{(\text{mol l}^{-1})^{c+d}}{(\text{mol l}^{-1})^{a+b}} = (\text{mol l}^{-1})^{(c+d)-(a+b)} \\ &= (\text{mol l}^{-1})^{\Delta n_g} \end{aligned}$$

$$\Delta n_g = n_p - n_r = 2 - 4 = -2$$

$$\begin{aligned} K_{eq} &= (\text{mol l}^{-1})^{-2} \\ &= \text{mol}^{-2} \text{ l}^2 \end{aligned}$$

QUESTION



Lalke

For the reaction, $\text{SO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$ if $K_p = K_c (\text{RT})^x$ where, the symbols have usual meaning, then the value of x is (assuming ideality)

1 -1

2 $-\frac{1}{2}$

3 $\frac{1}{2}$

4 1

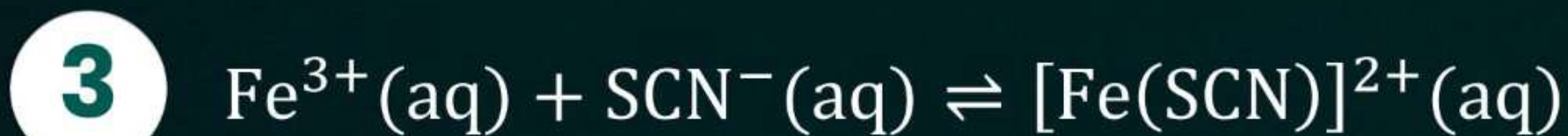
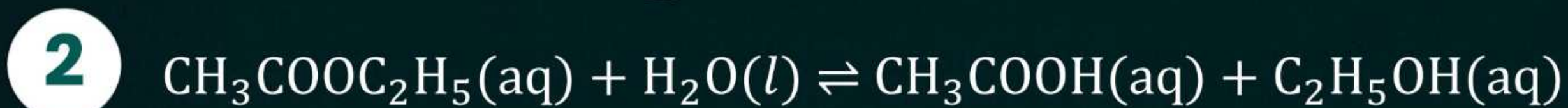
x has no value in my life.

$$\begin{aligned} x &= \Delta n_g = n_p - n_r \\ &= 1 - 1.5 \\ &= -\frac{1}{2} \text{ or } -0.5 \end{aligned}$$

QUESTION



Which of the following reactions is correct regarding homogeneous equilibria?

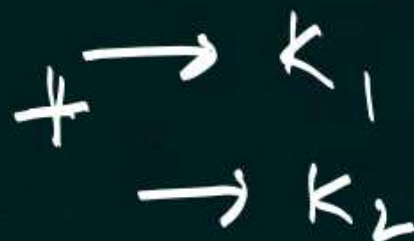


4 All of the above

QUESTION



The equilibrium constants for the reaction,



are K_1 and K_2 respectively. The equilibrium constant for the combined reaction is.

1

$$\frac{K_1}{K_2}$$

2

$$K_1 \times K_2$$

3

$$K_1 - K_2$$

4

$$\sqrt{K_1 \times K_2}$$

QUESTION

We know that the relationship between K_c is K_p is

$$K_p = K_c (RT)^{\Delta n}$$

What would be the value of Δn for the reaction?



$$\Delta n_g = n_p - n_r$$

$$= 2 - 0$$

$$= 2$$

1

1

2

0.5

3

1.5

4

2

EXAMPLE

The value of K_c for the reaction $2A \rightleftharpoons B + C$ is 2×10^{-3} . At a given time, the composition of reaction mixture is $[A] = [B] = [C] = 3 \times 10^{-4} \text{M}$. In which direction the reaction will proceed?

QUESTION

In which condition, the reaction proceeds in the forward direction?

1 $Q_c = K_c$

2 $Q_c > K_c$

3 $Q_c < K_c$

4 $Q_c \neq K_c$

EXAMPLE



$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \\ = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$$

For the equilibrium



The value of the equilibrium constant, K_c is 3.17×10^{-6} at 1069 K.

Calculate the K_p for the reaction at this temperature?

Sol. Given: $\rightarrow$

$$K_c = 3.17 \times 10^{-6}$$

$$T = 1069 \text{ K}$$

$$K_p = ?$$

We know that

$$K_p = K_c (RT)^{\Delta n_g}$$

$$= 3.17 \times 10^{-6} (8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 1069 \text{ K})^1$$

$$= 3.17 \times 10^{-6} \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 1069 \text{ K}$$

=

$$\left\{ \Delta n_g = n_p - n_R = 3 - 2 = 1 \right.$$

EXAMPLE

Which of the following reactions involve homogeneous equilibrium and which ones involve heterogeneous equilibrium?

- (i) $2\text{N}_2\text{O}(\text{g}) \rightleftharpoons 2\text{N}_2(\text{g}) + \text{O}_2(\text{g})$ (Homogeneous)
- (ii) $2\text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ (Homogeneous)
- (iii) $2\text{Cu}(\text{NO}_3)_2(\text{s}) \rightleftharpoons 2\text{CuO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ (Heterogeneous)
- (iv) $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ (Heterogeneous)
- (v) $\text{Fe}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightleftharpoons \text{Fe}(\text{OH})_3(\text{s})$ (Heterogeneous)

Homogeneous Equilibrium = Same physical state of Reactant & Product

Heterogeneous Equilibrium = diff. phys. state of Reactant & Product.

QUESTION

In the equilibrium, $AB \rightleftharpoons A + B$, if the equilibrium concentration of A is double, then equilibrium concentration of B will be

- 1** Half
- 2** Twice
- 3** $1/4^{\text{th}}$
- 4** $1/8^{\text{th}}$

$$K_{eq} = \frac{[A][B]}{[AB]}$$

$$K_{eq} = \frac{[2A] \cdot [\frac{1}{2}B]}{[AB]} =$$

Because K_{eq} does not depend upon conc. only temp. dependent.

- ① Mohd. Abbas.
- ② Sehrul Jannat
- ③ Sarfaraz Ahmed
Bahroo.

JIGAR. K Mayal Rakho.

Thank
You

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